

One-Step and Two-Step Equations

USA Grade 6 Math
CCSS 6.EE.B.5-7

Name: _____

Date: _____

LEARNING TARGET

I can solve equations and check solutions by substitution.

KEY STRATEGY + WORKED EXAMPLE

Undo operations in reverse order while keeping both sides balanced; substitute the solution into the original equation.

$3x + 5 = 26$; subtract 5: $3x=21$; divide by 3: $x=7$.

A. TRY TOGETHER - USE THE STRATEGY

1. Solve: $x + 7 = 19$; $x =$ _____

2. Solve: $3x = 42$; $x =$ _____

3. Solve: $x/5 = 9$; $x =$ _____

4. Solve: $x - 13 = 8$; $x =$ _____

B. CORE PRACTICE - SHOW YOUR WORK

5. Solve: $4x + 3 = 31$; $x =$ _____

6. Solve: $2x - 5 = 17$; $x =$ _____

7. Solve: $x/6 + 4 = 9$; $x =$ _____

8. Solve: $5(x-2)=40$; $x =$ _____

9. Solve: $3x+8=2$; $x =$ _____

10. Solve: $7-x=12$; $x =$ _____

11. Solve: $x + 7 = 19$; $x =$ _____

12. Solve: $3x = 42$; $x =$ _____

C. INDEPENDENT FLUENCY - CALCULATE AND CHECK

13. Solve: $x/5 = 9$; $x =$ _____

14. Solve: $x - 13 = 8$; $x =$ _____

15. Solve: $4x + 3 = 31$; $x =$ _____

16. Solve: $2x - 5 = 17$; $x =$ _____

17. Solve: $x/6 + 4 = 9$; $x =$ _____

18. Solve: $5(x-2)=40$; $x =$ _____

19. Solve: $3x+8=2$; $x =$ _____

20. Solve: $7-x=12$; $x =$ _____

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D. APPLY THE SKILL

21. A number plus 18 equals 47. Write and solve an equation.

22. Seven tickets cost \$52.50 equally. Write an equation for price per ticket.

23. A taxi fare is \$4 plus \$2.50 per mile and totals \$24. Write and solve.

E. REASON, CHECK AND CORRECT

24. Explain why the same operation must be performed on both sides.

25. Correct the work: $4x+6=30$, so $x=30-6=24$.

EXIT TICKET + CHALLENGE

26. Solve $6x-11=43$ and check by substitution.

Challenge: Create a two-step equation with solution -4 and show a balance-model solution.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Explain one strategy you used today.

28. Check one answer in a different way.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: _____

31. The most difficult part was: _____

Why was it difficult? _____

32. My next practice step is: _____

TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

- A. 1) 12, 2) 14, 3) 45, 4) 21
B. 5) 7, 6) 11, 7) 30, 8) 10, 9) -2, 10) -5, 11) 12, 12) 14
C. 13) 45, 14) 21, 15) 7, 16) 11, 17) 30, 18) 10, 19) -2, 20) -5
D. 21) $x+18=47$; $x=29$. 22) $7p=52.50$; $p=\$7.50$. 23) $4+2.5m=24$; $m=8$ miles.
E. 24) Equal operations preserve balance. 25) $4x=24$, so $x=6$.
EXIT. 26) $x=9$; $6(9)-11=43$.

Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.

F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may stop after the first inverse operation. Isolate the variable fully and check in the original equation.

TEACHER CHECK

- ☐ Uses an appropriate Grade 6 Math strategy.
- ☐ Shows or explains thinking clearly.
- ☐ Checks a response using evidence, a rule, or a second method.
- ☐ Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.